

BAHLA, Oman

WMO: 412630

Lat: 22.9122N

Lon: 57.2572E

Elev: 513

StdP: 95.31

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.4	(c) 12.9	(d) -9.4	(e) 1.8	(f) 29.0	(g) -7.1	(h) 2.2	(i) 27.8	(j) 10.2	(k) 22.6	(l) 8.8	(m) 22.7	(n) 0.9	(o) 0	(p) 0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 12.1	(c) 44.2	(d) 19.9	(e) 43.1	(f) 20.1	(g) 42.1	(h) 20.1	(i) 25.2	(j) 33.0	(k) 24.6	(l) 33.1	(m) 24.0	(n) 33.2	(o) 4.5	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 23.2	(b) 19.1	(c) 29.2	(d) 22.3	(e) 18.1	(f) 29.2	(g) 21.5	(h) 17.2	(i) 29.3	(j) 80.4	(k) 32.7	(l) 77.6	(m) 33.0	(n) 75.4	(o) 33.1	(p) 26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.2	(b) 7.7	(c) 6.3	(e) 9.0	(f) 46.1	(g) 1.3	(h) 0.7	(i) 8.0	(j) 46.7	(k) 7.3	(l) 47.1	(m) 6.5	(n) 47.5	(o) 5.5	(p) 48.0		
			DB													
			WB													
				3.6	25.6	1.9	0.9	2.3	26.2	1.2	26.8	0.2	27.3	-1.2	27.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	19.2	21.5	25.3	29.8	33.9	35.8	35.5	34.3	32.4	29.3	24.3	20.6
	DBStd	6.21	1.97	2.42	2.51	2.38	1.93	2.11	2.30	2.27	1.74	1.80	1.85	2.03
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	23	14	3	0	0	0	0	0	0	0	0	0	6
	CDD10.0	6762	286	322	473	595	742	773	789	754	673	598	428	329
	CDD18.3	3743	42	91	215	345	483	523	531	496	423	339	178	77
	CDH23.3	56076	274	819	2462	4927	8021	8973	8973	7973	6603	4739	1753	558
(14) Wind	WSAvg	2.5	2.0	2.4	2.6	2.7	3.1	3.2	2.8	2.6	2.5	2.3	1.9	1.8
	PrecAvg	118	8	17	15	18	8	8	16	19	7	7	5	2
	PrecMax	205	63	104	87	86	33	34	40	48	20	48	40	21
	PrecMin	45	0	4	0	0	0	0	5	0	0	0	0	0
	PrecStd	47	16	23	22	25	9	9	9	12	5	13	10	5
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.3	32.5	36.2	39.7	44.6	46.0	45.2	44.1	41.5	39.0	33.7	30.3
		MCWB	14.3	16.1	16.2	18.0	18.8	19.6	20.6	20.8	19.7	18.2	17.4	15.5
	2%	DB	27.4	30.7	34.5	38.1	42.3	44.5	43.8	42.8	40.3	37.6	32.2	28.8
		MCWB	14.5	15.6	16.0	17.6	19.0	19.4	20.4	20.7	19.5	17.8	16.6	15.1
	5%	DB	26.1	29.2	33.2	37.0	40.9	43.2	42.8	41.7	39.3	36.6	31.1	27.7
		MCWB	14.4	15.2	16.2	17.4	18.8	19.5	20.6	20.7	19.5	17.6	16.7	15.3
	10%	DB	24.8	27.8	31.9	35.9	39.8	42.1	41.7	40.5	38.3	35.4	30.0	26.4
		MCWB	14.2	15.0	15.9	16.9	18.6	19.6	20.7	21.1	19.5	17.4	16.3	15.2
	0.4%	WB	18.2	19.7	21.2	22.2	24.8	26.1	25.7	25.5	24.2	22.7	22.1	18.9
		MCDB	23.2	25.3	26.6	28.1	29.9	32.1	34.8	33.4	33.1	29.5	27.2	22.4
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	17.1	18.6	20.1	20.9	23.7	25.3	25.0	24.5	23.6	21.7	19.8	18.0
		MCDB	22.2	25.0	26.2	28.4	30.7	32.3	33.9	33.3	32.3	28.9	26.1	23.5
	5%	WB	16.4	17.7	19.2	20.0	23.0	24.5	24.5	24.0	23.1	20.9	18.8	17.4
		MCDB	22.1	24.8	26.8	29.5	31.5	32.6	33.7	34.0	31.6	29.1	26.0	23.6
	10%	WB	15.7	16.7	18.3	19.1	22.0	23.7	24.0	23.5	22.5	20.1	18.1	16.8
		MCDB	22.1	24.3	26.6	31.4	32.5	33.0	33.4	33.7	31.4	29.4	26.0	23.6
(36) Mean Daily Temperature Range	5% DB	MDBR	11.7	12.1	12.2	11.4	11.5	12.1	11.8	12.0	12.0	12.0	11.3	11.5
		MCDBR	12.9	13.6	13.4	12.1	12.6	13.2	12.4	12.7	12.5	12.8	12.0	12.9
		MCWBR	4.9	5.1	4.7	3.7	4.0	4.2	3.4	3.0	3.5	4.0	4.4	4.9
	5% WB	MCDBR	10.1	10.4	9.9	10.6	11.2	12.4	11.8	12.2	11.9	11.0	10.2	10.2
		MCWBR	4.2	4.4	4.2	3.9	4.8	4.5	2.9	2.7	3.2	3.7	3.9	4.3
(40) Clear-Sky Solar Irradiance	taub		0.424	0.480	0.556	0.626	0.662	0.741	0.816	0.777	0.658	0.516	0.453	0.421
	taud		2.087	1.918	1.733	1.627	1.568	1.467	1.396	1.443	1.601	1.854	2.016	2.088
	Ebn at Noon		826	802	761	717	688	631	585	607	673	758	788	811
	Edh at Noon		147	185	233	263	278	305	328	313	262	195	156	142
(44) All-Sky Solar Radiation	RadAvg		4.74	5.53	6.38	7.03	7.60	7.43	6.86	6.79	6.46	5.91	4.95	4.49
	RadStd		0.20	0.23	0.33	0.37	0.16	0.26	0.27	0.16	0.12	0.16	0.12	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		+0.42	+0.68	N/A	N/A	+0.28	+0.30	N/A	N/A	+96

Nomenclature: See separate page